



Restricted Circulation

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PACKAGING RESEARCH & DEVELOPMENT

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For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

i-Fos (Ibandronic Acid Film Coated Tablets 150mg)

COMPOSITION

Each film coated tablet contains
Ibandronate Sodium monohydrate equivalent to Ibandronic acid.....150 mg

DESCRIPTION

White to off white, oblong, biconvex, film coated tablet debossed with "L 71" on one side and plain on the other side.

PHARMACOLOGICAL CLASSIFICATION

Ibandronate sodium is a nitrogen-containing bisphosphonate that inhibits osteoclast-mediated bone resorption. The chemical name for ibandronate sodium is 3-(N-methyl-N-pentyl) amino-1-hydroxypropane-1,1dip - phosphonic acid, monosodium salt, monohydrate with the molecular formula C9H22NO7P2Na•H2O and a molecular weight of 359.24.

PHARMACOLOGICAL ACTION:

Pharmacodynamics

Pharmacotherapeutic group: Medicinal products for treatment of bone diseases, Bisphosphonates, ATC code: M05BA06

Mechanism of action

Ibandronic acid is a highly potent bisphosphonate belonging to the nitrogen-containing group of bisphosphonates, which act selectively on bone tissue and specifically inhibit osteoclast activity without directly affecting bone formation. It does not interfere with osteoclast recruitment. Ibandronic acid leads to progressive net gains in bone mass and a decreased incidence of fractures through the reduction of elevated bone turnover towards premenopausal levels in postmenopausal women. Daily or intermittent administration of ibandronic acid results in reduced bone resorption as reflected in reduced levels of serum and urinary biochemical markers of bone turnover, increased BMD and a decreased incidence of fractures.

Pharmacokinetics

The primary pharmacological effects of ibandronic acid on bone are not directly related to actual plasma concentrations, as demonstrated by various studies in animals and humans.

Absorption

The absorption of ibandronic acid in the upper gastrointestinal tract is rapid after oral administration and plasma concentrations increase in a dose-proportional manner up to 50 mg oral intake, with greater than doseproportional increases seen above this dose. Maximum observed plasma concentrations were reached within 0.5 to 2 hours (median 1 hour) in the fasted state and absolute bioavailability was about 0.6 %. The extent of absorption is impaired when taken together with food or beverages (other than water). Bioavailability is reduced by about 90 % when ibandronic acid is administered with a standard breakfast in comparison with bioavailability seen in fasted subjects. There is no meaningful reduction in bioavailability provided ibandronic acid is taken 60 minutes before the first food of the day. Both bioavailability and BMD gains are reduced when food or beverage is taken less than 60 minutes after ibandronic acid is ingested.

Distribution

After initial systemic exposure, ibandronic acid rapidly binds to bone or is excreted into urine. In humans, the apparent terminal volume of distribution is at least 90 l and the amount of dose reaching the bone is estimated to be 40-50 % of the circulating dose. Protein binding in human plasma is approximately 85 % - 87 % (determined *in vitro* at therapeutic concentrations), and thus there is a low potential for interaction with other medicinal products due to displacement.

Biotransformation

There is no evidence that ibandronic acid is metabolised in animals or humans.

Elimination

The absorbed fraction of ibandronic acid is removed from the circulation via bone absorption (estimated to be 40-50 % in postmenopausal women) and the remainder is eliminated unchanged by the kidney. The unabsorbed fraction of ibandronic acid is eliminated unchanged in the faeces. The range of observed apparent half-lives is broad, the apparent terminal half-life is generally in the range of 10-72 hours. As the values calculated are largely a function of the duration of study, the dose used, and assay sensitivity, the true terminal half-life is likely to be substantially longer, in common with other bisphosphonates. Early plasma levels fall quickly reaching 10 % of peak values within 3 and 8 hours after intravenous or oral administration respectively. Total clearance of ibandronic acid is low with average values in the range 84-160 ml/min. Renal clearance (about 60 mL/min in healthy postmenopausal females) accounts for 50-60 % of total clearance and is related to creatinine clearance. The difference between the apparent total and renal clearances is considered to reflect the uptake by bone. The secretory pathway appears not to include known acidic or basic transport systems involved in the excretion of other active substances. In addition, ibandronic acid does not inhibit the major human hepatic P450 isoenzymes and does not induce the hepatic cytochrome P450 system in rats.

Pharmacokinetics in special clinical situations

Gender

Bioavailability and pharmacokinetics of ibandronic acid are similar in men and women.

Race

There is no evidence for any clinically relevant inter-ethnic differences between Asians and Caucasians in ibandronic acid disposition. There are few data available on patients of African origin.

Renal impairment

Renal clearance of ibandronic acid in patients with various degrees of renal impairment is linearly related to creatinine clearance. No dose adjustment is necessary for patients with mild or moderate renal impairment (CL_{cr} equal or greater than 30 ml/min), as shown in study BM 16549 where the majority of patients had mild to moderate renal impairment. Subjects with severe renal failure (CL_{cr} less than 30 ml/min) receiving daily oral administration of 10 mg ibandronic acid for 21 days, had 2-3 fold higher plasma concentrations than subjects with normal renal function and total clearance of ibandronic acid was 44 ml/min. After intravenous administration of 0.5 mg, total, renal, and non-renal clearances decreased by 67 %, 77 % and 50 %, respectively, in subjects with severe renal failure but there was no reduction in tolerability associated with the increase in exposure. Due to the limited clinical experience, ibandronic acid is not recommended in patients with severe renal impairment. The pharmacokinetics of ibandronic acid was not assessed in patients with end-stage renal disease managed by other than haemodialysis. The pharmacokinetics of ibandronic acid in these patients is unknown, and ibandronic acid should not be used under these circumstances.

Hepatic impairment

There are no pharmacokinetic data for ibandronic acid in patients who have hepatic impairment. The liver has no significant role in the clearance of ibandronic acid which is not metabolised but is cleared by renal excretion and by uptake into bone. Therefore dose adjustment is not necessary in patients with hepatic impairment.

Elderly

In a multivariate analysis, age was not found to be an independent factor of any of the pharmacokinetic parameters studied. As renal function decreases with age this is the only factor to take into consideration.

Paediatric population

There are no data on the use of ibandronic acid in patients less than 18 years old.

INDICATIONS AND USAGE

i-Fos is indicated for:

Treatment of postmenopausal osteoporosis, to reduce the risk of fractures. Efficacy on femoral neck fractures has not been established.

Treatment of Osteoporosis: Osteoporosis may be confirmed by the finding of low bone mass (T score < - 2.0 SD) and the presence or history of osteoporotic fracture, or a low bone mass (T-score < - 2.5 SD) in the absence of documented pre-existing osteoporotic fracture.

DOSAGE AND ADMINISTRATION

The recommended dose of i-Fos for treatment is one 150 mg film-coated tablet once a month. The tablet should preferably be taken on the same date each month. i-Fos should be taken 60 minutes before the first food or drink (other than water) of the day (see section; Interaction with other Medicinal Products and other Forms of interaction- "Drug-Food Interaction") or any other oral medication or supplementation (including calcium); Tablets should be swallowed whole with a full glass of plain water (180 to 240 ml) while the patient is sitting or standing in an upright position. Patients should not lie down for 60 minutes after taking i-Fos. Plain water is the only drink that should be taken with i-Fos. Please note, that some mineral waters may have a higher concentration of calcium and therefore should not be used. Patients should not chew or suck the tablet because of a potential for oropharyngeal ulceration. Patients should receive supplemental calcium or vitamin D if dietary intake is inadequate. In case a once-monthly dose is missed, patients should be instructed to take one i-Fos 150 mg tablet the morning after the tablet is remembered, unless the time to the next scheduled dose is within 7 days. Patients should then return to taking their dose once a month on their originally scheduled date. If the next scheduled dose is within 7 days, patients should wait until their next dose and then continue taking one tablet once a month as originally scheduled. Patients should not take two 150 mg tablets within the same week.

Special Dosage Instructions

Patients with hepatic impairment

No dosage adjustment is necessary (see section; Pharmacokinetics in Special Populations).

Patients with renal impairment

No dosage adjustment is necessary for patients with mild or moderate renal impairment where creatinine clearance is ≥30 ml/min. Below 30 ml/min creatinine clearance, the decision to administer i-Fos should be based on an individual risk-benefit assessment (see section; Pharmacokinetics in Special Populations).

Elderly

No dosage adjustment is necessary.

Children

Safety and efficacy have not been established in patients less than 18 years old.

CONTRAINDICATIONS

- Hypersensitivity to ibandronic acid or to any of the excipients.

- Hypocalcaemia

- Abnormalities of the oesophagus which delay oesophageal emptying such as stricture or achalasia.

- Inability to stand or sit upright for at least 60 minutes.

WARNINGS & PRECAUTIONS

Hypocalcaemia

Existing hypocalcaemia must be corrected before starting Ibandronic acid therapy. Other disturbances of bone and mineral metabolism should also be effectively treated. Adequate intake of calcium and vitamin D is important in all patients.

Gastrointestinal irritation

Orally administered bisphosphonates may cause local irritation of the upper gastrointestinal mucosa. Because of these possible irritant effects and a potential for worsening of the underlying disease, caution should be used when Ibandronic acid is given to patients with active upper gastrointestinal problems (e.g. known Barrett's oesophagus, dysphagia, other oesophageal diseases, gastritis, duodenitis or ulcers). Adverse reactions such as oesophagitis, oesophageal ulcers and oesophageal erosions, in some cases severe and requiring hospitalisation, rarely with bleeding or followed by oesophageal stricture or perforation, have been reported in patients receiving treatment with oral bisphosphonates. The risk of severe oesophageal adverse experiences appears to be greater in patients who do not comply with the dosing instruction and/or who continue to take oral bisphosphonates after developing symptoms suggestive of oesophageal irritation. Patients should pay particular attention to and be able to comply with the dosing instructions. Physicians should be alert to any signs or symptoms signalling a possible oesophageal reaction and patients should be instructed to discontinue Ibandronic acid and seek medical attention if they develop dysphagia, odynophagia, retrosternal pain or new or worsening heartburn. While no increased risk was observed in controlled clinical trials there have been post-marketing reports of gastric and duodenal ulcers with oral bisphosphonate use, some severe and with complications. Since Nonsteroidal Anti-Inflammatory medicinal products and bisphosphonates are both associated with gastrointestinal irritation, caution should be taken during concomitant administration.



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Osteonecrosis of the external auditory canal
Osteonecrosis of the external auditory canal has been reported with bisphosphonates, mainly in association with long-term therapy. Possible risk factors for osteonecrosis of the external auditory canal include steroid use and chemotherapy and/or local risk factors such as infection or trauma. The possibility of osteonecrosis of the external auditory canal should be considered in patients receiving bisphosphonates who present with ear symptoms including chronic ear infections.

Osteonecrosis of the jaw
Osteonecrosis of the jaw (ONJ) has been reported in patients treated with bisphosphonates. Most cases have been in cancer patients undergoing dental procedures, but some have occurred in patients with postmenopausal osteoporosis or other diagnoses. Known risk factors for osteonecrosis of the jaw include a diagnosis of cancer, concomitant therapies (eg, chemotherapy, including angiogenesis inhibitors, radiotherapy, corticosteroids), and co-morbid disorders (eg, anemia, coagulopathy, infection, pre-existing dental disease). Most reported cases have been in patients treated with bisphosphonates intravenously but some have been in patients treated orally. For patients who develop osteonecrosis of the jaw while on bisphosphonate therapy, dental surgery may exacerbate the condition. For patients requiring dental procedures, there are no data available to suggest whether discontinuation of bisphosphonate treatment reduces the risk of ONJ. Clinical judgment of the treating physician should guide the management plan of each patient based on individual benefit/risk assessment.

Renal impairment
Due to limited clinical experience, ibandronic acid is not recommended for patients with a creatinine clearance below 30 ml/min.

Galactose intolerance
This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucosegalactose malabsorption should not take this medicinal product.

DRUG INTERACTIONS
Medicinal product-Food Interaction
Oral bioavailability of ibandronic acid is generally reduced in the presence of food. In particular, products containing calcium, including milk and other multivalent cations (such as aluminium, magnesium, iron), are likely to interfere with absorption of ibandronic acid, which is consistent with findings in animal studies. Therefore, patients should fast overnight (at least 6 hours) before taking ibandronic and continue fasting for 1 hour following intake of ibandronic acid.

Interactions with other medicinal products
Metabolic interactions are not considered likely, since ibandronic acid does not inhibit the major human hepatic P450 isoenzymes and has been shown not to induce the hepatic cytochrome P450 system in rats. Ibandronic acid is eliminated by renal excretion only and does not undergo any biotransformation.

Calcium supplements, antacids and some oral medicinal products containing multivalent cations
Calcium supplements, antacids and some oral medicinal products containing multivalent cations (such as aluminium, magnesium, iron) are likely to interfere with the absorption of ibandronic acid. Therefore, patients should not take other oral medicinal products for at least 6 hours before taking ibandronic acid and for 1 hour following intake of ibandronic acid.

Acetylsalicylic acid and NSAIDs
Since Acetylsalicylic acid, Nonsteroidal Anti-Inflammatory medicinal products (NSAIDs) and bisphosphonates are associated with gastrointestinal irritation, caution should be taken during concomitant administration.

H2 blockers or proton pump inhibitors
Of over 1500 patients enrolled in study BM 16549 comparing monthly with daily dosing regimens of ibandronic acid, 14 % and 18 % of patients used histamine (H2) blockers or proton pump inhibitors after one and two years, respectively. Among these patients, the incidence of upper gastrointestinal events in the patients treated with ibandronic acid 150 mg once monthly was similar to that in patients treated with ibandronic acid 2.5 mg daily. In healthy male volunteers and postmenopausal women, intravenous administration of ranitidine caused an increase in ibandronic acid bioavailability of about 20 %, probably as a result of reduced gastric acidity. However, since this increase is within the normal variability of the bioavailability of ibandronic acid, no dose adjustment is considered necessary when Ibandronic acid is administered with H2-antagonists or other active substances which increase gastric pH. Pharmacokinetic interaction studies in postmenopausal women have demonstrated the absence of any interaction potential with tamoxifen or hormone replacement therapy (estrogen). No interaction was observed when co-administered with melphalan/prednisolone in patients with multiple myeloma.

PREGNANCY & LACTATION:
Pregnancy
Ibandronic acid is only for use in postmenopausal women and must not be taken by women of childbearing potential. There are no adequate data from the use of ibandronic acid in pregnant women. Studies in rats have shown some reproductive toxicity. The potential risk for humans is unknown. Adverse effects of ibandronic acid in reproductive toxicity studies in the rat were those observed with biphosphonates as a class. They include a decreased number of implantation sites, interference with natural delivery (dystocia), and an increase in visceral variations (renal pelvis ureter syndrome). Ibandronic acid should not be used during pregnancy.

Breast-feeding
It is not known whether ibandronic acid is excreted in human milk. Studies in lactating rats have demonstrated the presence of low levels of ibandronic acid in the milk following intravenous administration. Ibandronic acid should not be used during breast-feeding.

Fertility
There are no data on the effects of ibandronic acid from humans. In reproductive studies in rats by the oral route, ibandronic acid decreased fertility. In studies in rats using the intravenous route, ibandronic acid decreased fertility at high daily doses.

ADVERSE REACTION
Summary of the safety profile
The most serious reported adverse reactions are anaphylactic reaction/shock, atypical fractures of the femur, osteonecrosis of the jaw, gastrointestinal irritation, ocular inflammation. The most frequently reported adverse reactions are arthralgia and influenza-like symptoms. These symptoms are typically in association with the first dose, generally of short duration, mild or moderate in intensity, and usually resolve during continuing treatment without requiring remedial measures.

Tabulated list of adverse reactions
Adverse reactions are listed according to MedDRA system organ class and frequency category. Frequency categories are defined using the following convention: very common (>1/10), common (≥ 1/100 to < 1/10), uncommon (≥ 1/1,000 to < 1/100), rare (≥ 1/10,000 to < 1/1,000), very rare (<1/10,000), not known (cannot be estimated from the available data). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1: Adverse reactions occurring in postmenopausal women receiving ibandronic acid 150mg once monthly or ibandronic acid 2.5 mg daily

System Organ Class	Common	Uncommon	Rare	Very rare
Immune system disorders		Asthma exacerbation	Hypersensitivity reaction	Anaphylactic reaction/shock*†
Nervous system disorders	Headache	Dizziness		
Eye disorders		Ocular inflammation*†		
Gastrointestinal disorders*	Oesophagitis, Gastritis, Gastroesophageal reflux disease, Dyspepsia, Diarrhoea, Abdominal pain, Nausea	Oesophagitis including oesophageal ulcerations or strictures and dysphagia, Vomiting, Flatulence		
Skin and subcutaneous tissues disorders	Rash		Angioedema, Face oedema, Urticaria	Steven-Johnson Syndrome†, Erythema Multiforme†, Dermatitis Bullous†
Musculoskeletal and connective tissue disorders	Arthralgia, Myalgia, Musculoskeletal pain, Muscle cramp, Musculoskeletal stiffness	Back pain	Atypical subtrochanteric and diaphyseal femoral fractures†	Osteonecrosis of jaw*†, osteonecrosis of the external auditory canal (bisphosphonate class adverse reaction)
General disorders and administration site conditions	Influenza like illness*	Fatigue		

*See further information below
†Identified in post marketing experience.

Description of selected adverse reactions
Gastrointestinal adverse reactions:
Patients with a previous history of gastrointestinal disease including patients with peptic ulcer without recent bleeding or hospitalisation, and patients with dyspepsia or reflux controlled by medication were included in the once monthly treatment study. For these patients, there was no difference in the incidence of upper gastrointestinal adverse events with the 150 mg once monthly regimen compared to the 2.5 mg daily regimen.

Influenza-like illness
Influenza-like illness includes events reported as acute phase reaction or symptoms including myalgia, arthralgia, fever, chills, fatigue, nausea, loss of appetite, or bone pain.

Osteonecrosis of jaw
Cases of osteonecrosis of the jaw have been reported, predominantly in cancer patients treated with medicinal products that inhibit bone resorption, such as ibandronic acid. Cases of ONJ have been reported in the post marketing setting for ibandronic acid.

Ocular inflammation
Ocular inflammation events such as uveitis, episcleritis and scleritis have been reported with ibandronic acid. In some cases, these events did not resolve until the ibandronic acid was discontinued.

Anaphylactic reaction/shock
Cases of anaphylactic reaction/shock, including fatal events, have been reported in patients treated with intravenous ibandronic acid.

OVERDOSE:
No specific information is available on the treatment of overdose with ibandronic acid.
However, based on a knowledge of this class of compounds, oral overdose may result in upper gastrointestinal adverse reactions (such as upset stomach, dyspepsia, oesophagitis, gastritis, or ulcer) or hypocalcaemia. Milk or antacids should be given to bind ibandronic acid, and any adverse reactions treated symptomatically. Owing to the risk of oesophageal irritation, vomiting should not be induced and the patient should remain fully upright.

STORAGE
Store below 30°C in a dry place.

PRESENTATION
Alu Clear PVC/PE/PVDC Blister Pack of 1 Tablet

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